

Cloud Computing & DevOps – AWS Handbook

SESSION 6A

Infrastructure as Code (IaC) with AWS CloudFormation

Automating AWS Infrastructure Deployment

Duration	2 Hours
Prerequisites	AWS Account, EC2, S3 and Basic VPC Knowledge
Learning Outcome	Create, update and manage AWS infrastructure using CloudFormation templates and Infrastructure as Code principles.

Why Are We Learning This?

Infrastructure as Code allows cloud resources to be provisioned through code rather than manually through the AWS Console. AWS CloudFormation enables repeatable, automated and version-controlled deployments.

Architecture Diagram



Key Concepts

Infrastructure as Code	Managing infrastructure through code
Template	YAML or JSON definition file
Stack	Collection of AWS resources
Resource	Individual AWS component
Change Set	Preview before deployment
Nested Stack	Reusable stack component

Practical Activities

- Create a YAML CloudFormation template
- Deploy an S3 bucket stack
- Verify resource creation
- Update an existing stack
- Deploy an EC2 instance through CloudFormation
- Explore nested stacks
- Delete stacks and resources

Expected Outputs

Stack created successfully, S3 bucket deployed, EC2 instance launched and stack updates completed.

What You Should See

Paste Screenshot Here

CloudFormation Dashboard

Paste Screenshot Here

Stack Creation Status

Paste Screenshot Here

S3 Bucket Created by Stack

Paste Screenshot Here

EC2 Instance Created by Stack

Paste Screenshot Here

Nested Stack Configuration

Journal Writing Template

Aim

To provision AWS infrastructure using AWS CloudFormation.

Theory

CloudFormation automates AWS resource deployment using templates.

Procedure

Write 5–7 concise steps performed.

Observations

Resources were provisioned successfully using templates.

Conclusion

Infrastructure deployment was automated using CloudFormation.

Viva Preparation

1. What is Infrastructure as Code?
2. What is AWS CloudFormation?
3. What is a Template?
4. What is a Stack?
5. What is a Resource?
6. What is a Change Set?
7. What is a Nested Stack?
8. Advantages of IaC?

Troubleshooting Box

Problem	Possible Cause
Stack creation failed	Template syntax error
Resource not created	Permission issue
EC2 launch failure	Invalid AMI or subnet
Nested stack failure	Template URL issue

Industry Relevance

CloudFormation is a core AWS DevOps service used for automated infrastructure provisioning and CI/CD integration.

Session Summary

Students learned how to create, update and manage AWS infrastructure using Infrastructure as Code principles.